WoSM Academy

CS102-F26 · DIGITAL DESIGN AND COMPUTER ARCHITECTURE I

Flip-Flops

Lecture 12: edge-triggered memory: D, T and JK flip-flops

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Today

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02 The master–slave D flip-flop

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01

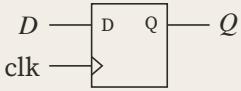
Edges instead of levels

Latches and flip-flops

- A latch reacts to a level: while $E = 1$ it is transparent. That is what went wrong at the end of Lecture 11.
- A flip-flop reacts to an edge: the instant the clock goes from 0 to 1 (or from 1 to 0).

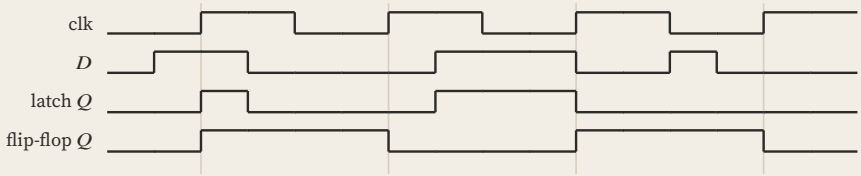
Definition (D flip-flop)

A rising-edge *D flip-flop* stores the value of D at each rising edge of the clock, and keeps it until the next one.



The triangle marks an edge-triggered clock input.

A latch and a flip-flop, side by side

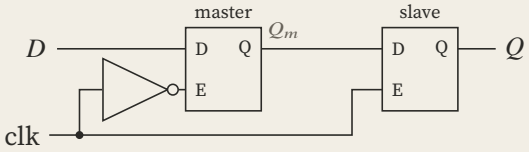


- The latch, enabled by the clock, follows every change of D while the clock is 1.
- The flip-flop changes only at the rising edges (the grey lines), to the value D had just before the edge, and holds it for a whole period.

02

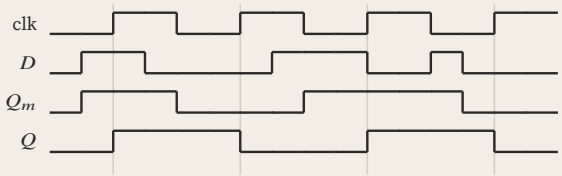
The master–slave D flip-flop

Two latches make a flip-flop



- The master latch is enabled by clk′: it is transparent while the clock is 0.
- The slave latch is enabled by clk: it is transparent while the clock is 1.
- The two are never transparent at the same time, so D never runs straight through to Q.

How it works



- Clock 0: the master follows D; the slave holds Q.
- Rising edge: the master closes on the last value of D; the slave opens and passes it to Q.
- Clock 1: D may change, but the master is closed.
- Falling edge: the slave closes; the master opens again.

Together: Q takes the value of D at each rising edge. In one line, $Q^+ = D$.

03

Reset

A known starting state

- When a circuit is switched on, its flip-flops hold random values. A reset input puts them in a known state, $Q = 0$.
- *Synchronous* reset: takes effect at the next clock edge. Just feed $D \cdot R'$ into the flip-flop.
- *Asynchronous* reset: takes effect at once, whatever the clock is doing.

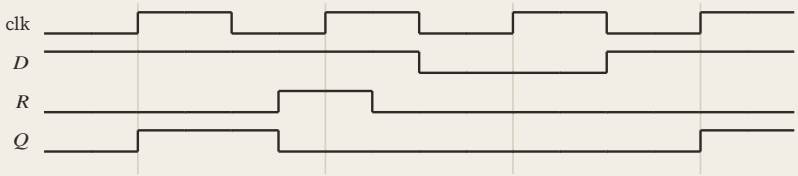
R	clk	D	Q
0	0	d	holds
0	↑	d	d
0	1	d	holds
0	↓	d	holds
1	any	any	0

The book's asynchronous reset adds a few gates in front of both latches:

- master: data $D \cdot R'$, enable $\text{clk}' + R$;
- slave: data $Q_m \cdot R'$, enable $\text{clk} + R$.

With $R = 1$ both latches are open and both receive 0: $Q = 0$ at once.

Asynchronous reset in time



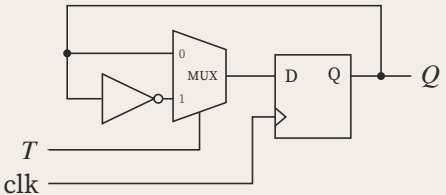
- R rises between two edges, and Q drops to 0 straight away.
- The edge at $t = 6$ comes while $R = 1$: it is ignored.
- After R falls, the flip-flop works normally again: the edges at 10 and 14 store 0 and 1.

The DX8's registers all have a reset input, so the whole computer starts from zero (Lecture 01's diagram).

04

T and JK flip-flops

The T flip-flop



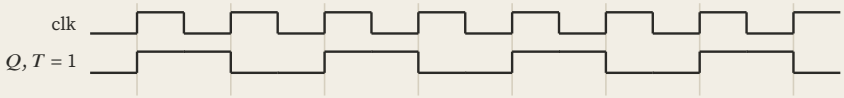
T	Q^+
0	Q (hold)
1	Q' (toggle)

$Q^+ = T \oplus Q$, at each rising edge.

Built as in the book: a D flip-flop, and a multiplexer that feeds it Q or Q'.

Lecture 01's lamp with a push button is a T flip-flop, and each press toggles it.

Toggling halves the frequency



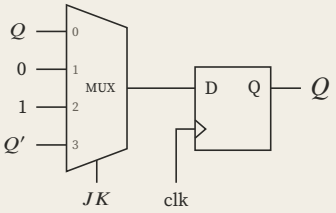
- With $T = 1$, Q flips at every rising edge: one period of Q takes two clock periods.
- A 100 MHz clock gives a 50 MHz Q. Chain such flip-flops, and each one halves the frequency again: that is a counter `(Unit 13)`.

The JK flip-flop

J	K	Q^+
0	0	Q (hold)
0	1	0 (reset)
1	0	1 (set)
1	1	Q' (toggle)

at each rising edge; between edges, Q holds

- Like an SR latch with J for set and K for reset, but $J = K = 1$ is allowed: it toggles.
- As an equation: $Q^+ = JQ' + K'Q$.
- Built as in the book: a D flip-flop fed by a 4-to-1 multiplexer with select JK and data Q, 0, 1, Q'.



All of them

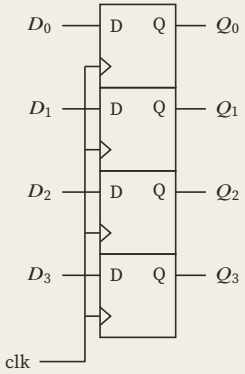
	Reacts to	Next value	Watch out
SR latch	levels	$Q^+ = S + R'Q$	$S = R = 1$ not allowed
D latch	the level of E	$Q^+ = D$ while $E = 1$	transparent
D flip-flop	clock edges	$Q^+ = D$	
T flip-flop	clock edges	$Q^+ = T \oplus Q$	
JK flip-flop	clock edges	$Q^+ = JQ' + K'Q$	

> **Remark**
>
> The D flip-flop is enough to build the other two, as we just did. Most digital designs, including the DX8, use D flip-flops everywhere.

05

Registers

From one bit to a register



- n D flip-flops that share one clock store an n-bit number: a register. The book's 8-bit register is just 8 of them.
- To load only when told, put a 2-to-1 multiplexer in front of each D: it passes Q_i (keep) or D_i (load), chosen by a write-enable bit.
- The DX8's register file holds 8 such registers `(Unit 19)`.

Try it

1. A D flip-flop starts with $Q = 0$. Just before four rising edges, D is 1, 0, 0, 1. What is Q after each edge?

2. A T flip-flop with $T = 1$ runs on a 100 MHz clock. What does its output look like?

3. A JK flip-flop has $J = K = 1$ and starts at $Q = 0$. What is Q after three rising edges?

4. Build a T flip-flop and a D flip-flop from a JK flip-flop.

Try it

1. A D flip-flop starts with $Q = 0$. Just before four rising edges, D is 1, 0, 0, 1. What is Q after each edge?
   1, 0, 0, 1: it copies D at each edge.
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   1, 0, 0, 1: it copies D at each edge.
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   A 50 MHz square wave: it flips once per clock period.
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3. A JK flip-flop has $J = K = 1$ and starts at $Q = 0$. What is Q after three rising edges?
   1, then 0, then 1: it toggles each time.
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   A 50 MHz square wave: it flips once per clock period.
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   1, then 0, then 1: it toggles each time.
4. Build a T flip-flop and a D flip-flop from a JK flip-flop.
   T: $J = K = T$. D: $J = D$, $K = D'$.

06

Summary

Summary

- A flip-flop changes only at a clock edge; a latch is transparent for as long as it is enabled.
- The master–slave D flip-flop: two D latches enabled by clk′ and clk. Q takes the value of D at each rising edge.
- Reset puts a flip-flop in a known state: synchronously at the next edge, or asynchronously at once.
- T flip-flop: $Q^+ = T \oplus Q$. JK flip-flop: $Q^+ = JQ' + K'Q$. Both are a D flip-flop with a multiplexer in front.
- n flip-flops sharing a clock form an n-bit register.

Before the next lecture

- Read Sections 5.3.3, 5.3.4 and 5.3.5 of the book.
- In Logisim-evolution, build the master–slave D flip-flop from your D latches, add the asynchronous reset, and compare it with the D flip-flop in the Memory library.
- Build a T flip-flop from it, connect T to 1, and watch Q in a timing diagram.

Next lecture (Unit 13)

Counters: the synchronous up counter and the asynchronous ripple counter.